THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL UNIT TEST SET 1

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The square of the distance between $(1, 4)$ and $(-3, 2)$ is: 1

(a) 17 **(b)** 20 **(c)** 25 **(d)** 22

2. If $x = 2$ is a zero of $7x + m$, then $m =$ 1

(a) -14 **(b)** -12 **(c)** -15 **(d)** -13

3. Which of the following has a terminating decimal expansion? 1

(a) $\frac{3}{7}$ **(b)** $\frac{39}{65}$ **(c)** $\frac{5}{18}$ **(d)** $\frac{14}{21}$

4. $(7x + 4y)^2 =$ 1

(a) $49x^2 + 56xy + 16y^2$ **(b)** $49x^2 + 28xy + 16y^2$ **(c)** $49x^2 + 16y^2$ **(d)** $49x^2 - 56xy + 16y^2$

5. The midpoint of $(3, 2)$ and $(6, 7)$ is: 1

(a) $(4.5,\ 5.5)$ **(b)** $(4.5,\ 4.5)$ **(c)** $(-1.5,\ -2.5)$ **(d)** $(5,\ 4.5)$

6. The slope of the line through $(1, 3)$ and $(3, 9)$ is: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 5

7. The number of decimal places in the decimal expansion of $\frac{7}{400}$ is: 1

(a) 6 **(b)** 3 **(c)** 5 **(d)** 4

8. $(2x + 3)(3x + 5) =$ 1

(a) $6x^2 + 19x + 15$ **(b)** $6x^2 + 20x + 15$ **(c)** $6x^2 + x + 15$ **(d)** $6x^2 + 19x - 15$

9. One end of a diameter of a circle is $(5, 1)$ and the centre is $(2, -1)$. The other end is: 1

(a) $(-3, -1)$ **(b)** $(-1, -3)$ **(c)** $(-1, 3)$ **(d)** $(1, -3)$

10. The x-intercept of the line $y = 4x + 12$ is: 1

(a) -4 **(b)** -1 **(c)** -3 **(d)** -2

11. $0.\overline{12}$ in lowest terms is: 1

(a) $\frac{2}{17}$ **(b)** $\frac{4}{33}$ **(c)** $\frac{1}{11}$ **(d)** $\frac{5}{33}$

12. If $a - b = 3$ and $ab = 18$, then $a^2 + b^2 =$ 1

(a) 46 **(b)** 45 **(c)** 44 **(d)** 47

13. The point on the x-axis equidistant from $(1, 5)$ and $(5, -5)$ is: 1

(a) $(3, 0)$ **(b)** $(0, 3)$ **(c)** $(-3, 0)$ **(d)** $(4, 0)$

14. The point $(k, 31)$ lies on $y = 7x + 3$. Then $k =$ 1

(a) 3 **(b)** 4 **(c)** 5 **(d)** 6

15. The length of the repeating block in the decimal expansion of $\frac{1}{7}$ is: 1

(a) 6 **(b)** 8 **(c)** 7 **(d)** 5

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the distance between $(1, 1)$ and $(4, 5)$. 2

17. Find the slope of the line through $(2, 4)$ and $(5, 13)$. 2

18. Express $0.8\overline{3}$ as a fraction. 2

19. Expand $(2x + 5)(3x + 1)$. 2

20. Find the midpoint of the segment joining $(3, -4)$ and $(6, 7)$. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Show that $A(1, 1)$, $B(4, 5)$, $C(10, 13)$ are collinear. 3

22. The line $y = mx + 12$ passes through $(2, 18)$. Find m, then the x-intercept of the line. 3

23. Express $0.\overline{027}$ as a fraction in lowest terms. 3

24. If $x + \frac{1}{x} = 6$, find $x^2 + \frac{1}{x^2}$ and $\left(x - \frac{1}{x}\right)^2$. 3

OR

Find the values of k for which the distance between $P(k, 1)$ and $Q(2, 5)$ is 5 units. 3

25. Find three points on the line $y = -4x + 5$ and show that the slope between the first and last is -4. 3

OR

Express $1.2\overline{3}$ in the form $\frac{p}{q}$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. The vertices of a rhombus are $A(-2, 2)$, $B(1, -2)$, $C(4, 2)$, $D(1, 6)$. Show that all four sides are equal and the diagonals are unequal, so that it is a rhombus and not a square. 5

OR

A linear polynomial $p(x)$ has $p(1) = 8$ and $p(3) = 16$. Find $p(x)$, its zero and $p(5)$. 5

27. Express $1.1\overline{6}$ as $\frac{p}{q}$ and verify by long division of p by q. 5

OR

If $x + y = 12$ and $x - y = 4$, find $x^2 - y^2$, xy and $x^2 + y^2$. 5

END OF QUESTION PAPER — **Total: 50 Marks**

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